

FPT UNIVERSITY

Course: ADY201m



PROJECT REPORT

REAL ESTATE PRICE ANALYSIS AND PREDICTION

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1. Introduction

The real estate market plays a pillar role in the economy but is often subjected to complex, multi-dimensional impacts from factors such as location, legal status, and amenities, making the determination of true property value a major challenge. This project approaches the problem by applying data science to exploit the rich source of information from real estate listings. The core objective is to quantify the impact of key attributes and build a reliable price prediction model. The research results are expected to provide an objective reference base, supporting effective decision-making for market participants and contributing to overall transparency.

Research Question: *"How can data analysis and machine learning tools be applied to interpret influencing factors and accurately predict real estate prices based on spatial characteristics and amenities?"*

1.1. Analysis and Prediction Objectives

Analysis: Focus on exploring the relationship between prices and attributes (area, location, amenities, legal status). The team evaluates the trend of price distribution by region and property type, while identifying potential legal risks from listing data.

Prediction: Build a price forecasting model based on user input information. The results provide a reference price to compare with the asking price, helping buyers negotiate more effectively.

2. Data Collection

To ensure a rich and multi-dimensional database for the analytical model, the research team aggregated data from two main sources: market transaction information and spatial data on urban infrastructure.

2.1. Real Estate Data

The scope of data collection focuses on the 4 most vibrant markets in the country, including: **Hanoi, Da Nang, Ho Chi Minh City, and Binh Duong**. Selecting these areas ensures a sufficiently large sample size and accurately reflects general market trends. The team utilized automated data collection techniques (Web Crawling) on the platform *batdongsan.com.vn* to extract important attributes such as selling price, area, house structure, and legal status.

2.2. Geographical and Amenity Data

Besides listing information, the location factor is quantified through the open map data **OpenStreetMap (OSM)**. The team applied spatial calculation algorithms to compile statistics on the density of essential amenities (schools, healthcare, commercial centers) within a defined radius around each property, thereby enriching the input information for the prediction model.

3. Overview of Raw Data

After the collection process, the raw dataset includes a total of **39 data fields**. Due to the large number of variables, the team categorized them into 4 main groups for convenient management before processing:

Data Group	No. of Vars	Description of Representative Variables
1. Listing Metadata	10 vars	Listing management info: <i>post_id</i> , <i>url</i> , <i>published_date</i> , <i>seller_name</i> ... helps identify and filter spam listings.
2. Geographical Location	9 vars	Administrative levels (<i>province</i> , <i>district</i> , <i>ward</i> ...) and coordinates (<i>latitude</i> , <i>longitude</i>).
3. Property Characteristics	14 vars	Asset information: <i>price_total</i> , <i>area</i> , <i>num_floors</i> , <i>num_bedrooms</i> , <i>direction</i> ...
4. External Amenities	6 vars	Number of amenities from OSM: <i>num_schools_1km</i> , <i>num_hospitals_2km</i> ...

Table 1: Overview of raw data structure (39 columns)

4. Methodology & Tools

4.1. Workflow

The project strictly adheres to the OSEM Data Science process: **Obtain** → **Scrub** → **Explore** → **Model** → **iNterpret**.

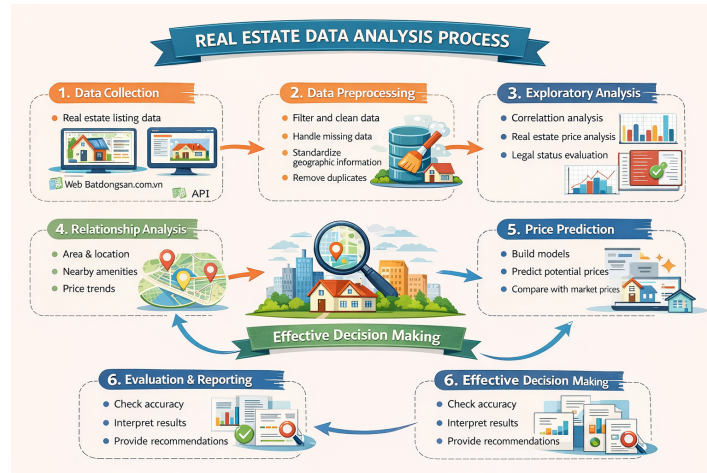


Figure 1: OSEM Data Analysis Process

4.2. Tools and Libraries

To ensure computational performance and visualization according to the above process, the team uses the following ecosystem:

- **Database:** Relational Database (SQL) used to manage, store, and extract data.
- **Processing & Analysis (Python):** pandas (Dataframe manipulation), numpy (numerical computation).
- **Visualization (Python):** matplotlib, seaborn (plotting EDA charts).
- **Machine Learning:** scikit-learn, catboost, xgboost, lightgbm (data preprocessing, model building, and training).
- **Deployment:** Streamlit (Web interface creation) and Folium (map display).

5. Data Processing Methods

Raw data collected from online environments often contains a lot of noise, inconsistent formats, and missing values. Based on the tools mentioned in the previous section, the team conducted a strict

preprocessing procedure to prepare inputs for the model.

5.1. Data Cleaning and Standardization

The processing procedure focuses on resolving formatting and consistency issues:

- **Format Standardization:** Quantitative attributes (price, area) were extracted and converted from text format to numerical format.
- **Data Synchronization:** Standardized administrative location information (District/County) into a unified standard category, removing incorrect name variations.
- **Deduplication:** Identified and removed duplicate listings to ensure data uniqueness.
- **Data Range Filtering:** Removed outliers, keeping only properties priced from **500 million to 60 billion VND**, with an area ranging from **15m² to 1000m²**.

5.2. Missing Data Handling and Feature Selection

From the cleaned dataset, the team further filtered:

- **Handling Missing Values:** Applied flexible imputation strategies based on the specific characteristics and distribution of each property type.
- **Feature Selection:** Removed variables that held no predictive value or caused model noise. The final result retained the **17 most important feature attributes** (as shown in Table 2).

Variable Group	No.	Variable Name	Type	Usage Meaning
1. Target	1	price_total	Int	Target variable: Total amount (VND).
	2	price_per_m2	Float	Reference unit price (VND/m ²).
2. Location	3	province	Category	Province/City.
	4	district	Category	District.
	5	latitude	Float	Latitude.
	6	longitude	Float	Longitude.
3. Property Characteristics	7	area	Float	Usable area (m ²).
	8	category	Category	Property type (Apartment, etc.).
	9	legal_status	Category	Legal status (Red book, etc.).
	10	frontage	Float	Frontage width (m).
	11	road_width	Float	Access road width (m).
	12	num_bedrooms	Int	Number of bedrooms.
	13	num_toilets	Int	Number of toilets.
	14	num_floors	Int	Number of floors.
4. Amenities	15	num_schools_1km	Int	School density (1km radius).
	16	num_hospitals_2km	Int	Healthcare density (2km radius).
	17	num_markets_1km	Int	Market density (1km radius).

Table 2: List of 17 feature variables categorized by logical groups

6. Exploratory Data Analysis (EDA)

6.1. Property Value Distribution & Correlation with Area

Through the total property value distribution chart, it can be seen that the data exhibits a clearly right-skewed distribution. The vast majority of listings are heavily concentrated in the price segment from 1 to 10 billion VND. The number of properties gradually decreases as the price increases. Premium segment assets account for a very small proportion.

Area is the feature with the largest impact weight. The scatter plot shows a clear positive correlation: The larger the area, the higher the total asset value. The team filtered out outliers (extremely large area but overly cheap price, or vice versa) to ensure input quality for the model.

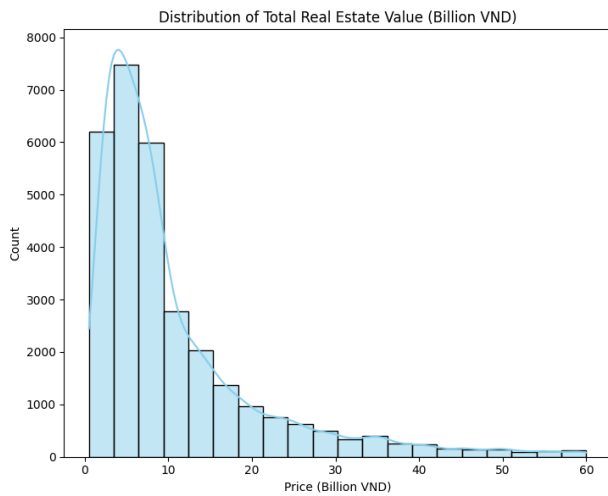


Figure 2: Distribution of Total Value (Billion VND)

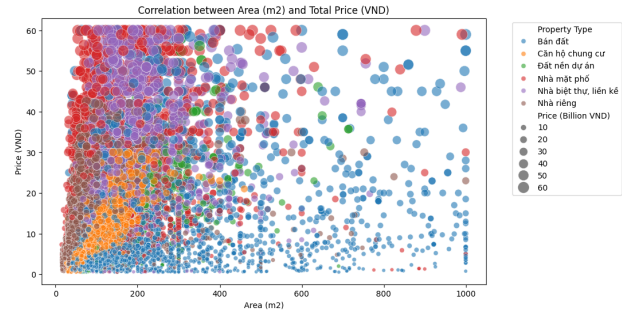


Figure 3: Correlation between Area and Price

6.2. Value Differentiation by Region

The data shows a clear differentiation in price levels among the 4 regions. Ho Chi Minh City and Hanoi have the highest average prices and large fluctuation margins. Conversely, Binh Duong and Da Nang are concentrated in a more stable price segment. This confirms that province and district are extremely important categorical variables in the model.

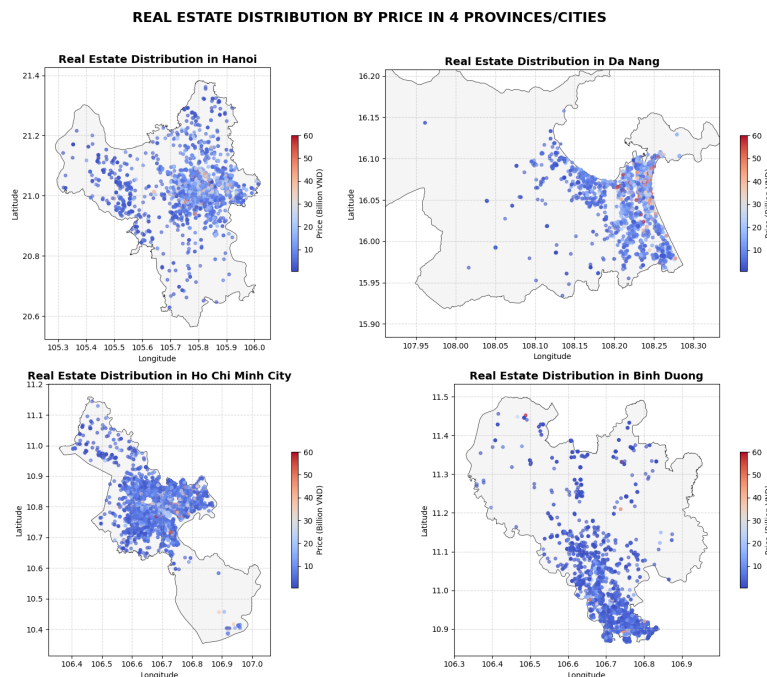


Figure 4: Average Price Segmented by Region

6.3. Correlation Matrix between Variables

To quantify the relationship between numerical features, the team used a Correlation Matrix (Heatmap). This matrix helps identify which factors have the strongest impact on real estate value.

The indices on the matrix show that total price (`price_total`) has a distinct positive correlation with three main factors: Area (`area` - 0.32), Number of floors (`num_floors` - 0.33), and Market/Supermarket density (`num_markets_1km` - 0.31). This result reflects market reality: large-scale houses located near bustling trading areas will have higher values. In addition, the concentration of utility infrastructure (schools and hospitals) has an extremely strong correlation (0.80), aiding the machine learning model in easily recognizing patterns.

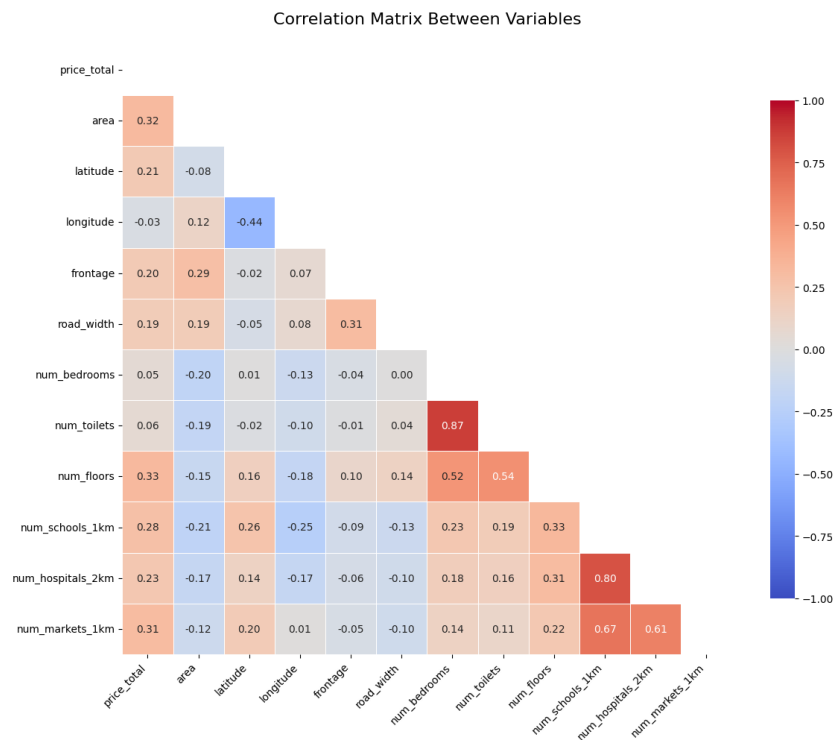


Figure 5: Correlation Matrix (Heatmap) between 12 numerical features

7. Model Building and Training

7.1. Preprocessing Pipeline

The team built an automated Pipeline: Imputing missing values, encoding categorical variables (One-Hot Encoder), and scaling quantitative variables (StandardScaler). The data was divided into Training and Testing sets at an **80:20** ratio.

7.2. Regression Analysis

Before building complex forecasting models, the team used **Ridge Regression** to evaluate the linear impact of each feature on house prices (Figure 6). The results showed:

- **Confirming patterns:** District variables (`district`) completely dominated the weight rankings. Central districts (Tay Ho, Ba Dinh, District 1) carry very large positive coefficients (strongly increasing prices), while suburban districts (Ba Vi, Cu Chi) carry large negative coefficients (decreasing prices).

- **Property Type:** The variable `category_Villas`, `Townhouses` is the only non-location factor to reach the top, showing that this premium property type adds immense value to an asset.
- **Limitations of linear models:** Although Ridge explains market trends well, the overall accuracy (R^2) only reached **0.57**. The interaction between location, amenities, and area in reality is non-linear, requiring more powerful machine learning algorithms for accurate forecasting.

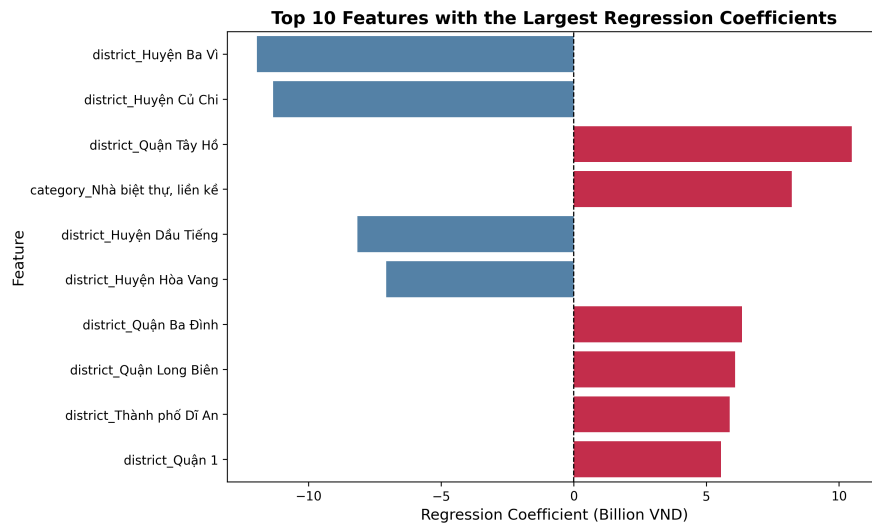


Figure 6: Top 10 Factors with the Largest Regression Weights Impacting House Prices

7.3. Machine Learning Model Evaluation and Selection

The team empirically trained 15 different Machine Learning algorithms (Ensemble, Tree-based, Boosting, etc.).

MODEL PERFORMANCE COMPARISON TABLE (SORTED FROM BEST TO WORST):				
	Model	R-squared (R2)	RMSE (Billion VND)	MAE (Billion VND)
0	Extra Trees	0.848608	4.016815	2.089744
1	CatBoost	0.845282	4.060707	2.322291
2	Random Forest	0.843933	4.078370	2.157832
3	XGBoost	0.843047	4.089936	2.326198
4	LightGBM	0.831225	4.241160	2.418388
5	Neural Network (MLP)	0.768947	4.962345	2.774025
6	Gradient Boosting	0.762728	5.028680	3.036464
7	SVR (Support Vector)	0.724651	5.417171	2.898606
8	KNN Regressor	0.695713	5.694725	3.076254
9	Decision Tree	0.649089	6.115467	3.045548
10	Linear Regression	0.573204	6.744370	4.512717
11	Ridge Regression	0.572382	6.750859	4.518957
12	Lasso Regression	0.505635	7.258633	4.770522
13	AdaBoost	0.504767	7.264999	5.524347
14	ElasticNet	0.498149	7.313385	4.743450

Figure 7: Performance Ranking of 15 Machine Learning Models

Decision: Based on the results table, **Extra Trees (Extremely Randomized Trees)** was the leading model with the highest R^2 score (0.84) and lowest error. This algorithm provides high accuracy by

combining multiple decision trees, minimizing overfitting through a random split selection mechanism. The team decided to select Extra Trees as the core model for deployment.

8. Discussion and Limitations

Practical Significance: The model successfully quantified the value of "external amenities" (schools, supermarkets), thereby affirming the importance of infrastructure planning on real estate values. The prediction results assist buyers and sellers in referencing objective prices, reducing information asymmetry in the market.

Limitations & Future Directions: Data from listings is often the **asking price** (seller's expectation), which may differ from the actual final transaction price on contracts. Additionally, the model has not yet captured "soft" variables such as interior design quality, Feng Shui orientation, or developer reputation. Future development directions will involve combining Natural Language Processing (NLP) techniques to analyze the "text description" sections of listings.

9. Practical Application Deployment

To actualize the model's value, the team used the **Streamlit** framework to build an interactive Web application. The interface integrates a smart map (Folium). When users click to select a location, the system automatically calls APIs (*Nominatim & Overpass*) to scan coordinates and count amenities. Combined with the Extra Trees model in the backend, the application instantly returns an estimated price (Billion VND) with an intuitive experience.

Figure 8: Web App Interface Integrating AI Real Estate Valuation Map

10. Conclusion

The project successfully executed a complete Data Science lifecycle, from data collection and analysis to Machine Learning model building. Successfully fine-tuning the Extra Trees algorithm provided good explanatory power for the problem of house price fluctuations. The outcome of the smart valuation Web App not only answers the research question well but also proves the tremendous practical value of artificial intelligence in the economic and real estate sectors.